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PI Name:	Ramachandran, Narayanan Ph.D.		
Project Title:	STUDY OF FLUID FLOW CONTROL IN PROTEIN CRYSTALLIZATION USING STRONG MAGNETIC FIELDS		
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Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Dynamics and stability		
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Zip Code:	35806	**Congressional District:**	5
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Task Description:	An important component in biotechnology, particularly in the area of protein engineering and rational drug design is the knowledge of the precise three-dimensional molecular structure of proteins. The quality of structural information obtained from X-ray diffraction methods is directly dependent on the degree of perfection of the protein crystals. As a consequence, the growth of high quality macromolecular crystals for diffraction analyses has been the central focus for biochemists, biologists, and bioengineers. To date, space protein crystallization experiments have yielded inconclusive results with ~20% of flight-grown proteins showing promising results. The space crystal growth environment offers reduced fluid motion – convection due solutal buoyancy and sedimentation effects. We postulate that we can reduce convection in ground-based experiments and we can understand the different aspects of convection control through the use of strong magnetic fields and field gradients. In 2004, progress was made on several fronts in the project. A key finding based on our measurements of the density and susceptibility variation of protein solutions as a function of protein concentration showed that an equivalent g-level of only ~0.2 go (where go is terrestrial gravity) is realizable in a practical, high field strength superconducting magnet. This is because the protein solution is weakly diamagnetic (repelled by a magnetic field) and requires a large magnetic field gradient to affect the solutal buoyancy driven flow. We developed a novel flow control methodology wherein we make the protein solution paramagnetic (attracted to a magnetic field) by adding paramagnetic salts. Fairly high paramagnetism can be induced by adding small quantities of the salt. We successfully grew protein crystals from these solutions and developed the theory for magnetic flow control. Validation experiments are planned for early 2005. A major advantage of the technique is that we can now do the magnetic flow control experiments in-house at NASA Marshall Space Flight Center using the available 5 Tesla magnet. The research on the project has resulted in several presentations, papers, and awards. We also developed key collaborations in applying our research to artificial gravity applications for supporting NASA's extended duration exploration missions. The concept is to use magnetic levitation as a microgravity model for cell cultures in order to determine the gravity threshold levels and artificial gravity requirements for exploration purposes. Task Objectives Specific goals for the investigation are: 1. To develop an understanding of convection control in diamagnetic fluids with concentration gradients through careful experimentation. Specifically solutal buoyancy driven convection due to crystal growth will be considered. Strong magnetic fields and field gradients will be utilized to unambiguously resolve the role of fluid flow in the process. 2. To develop predictive measures for successful crystallization in a magnetic field using analyses and numerical modeling for use in future protein crystal growth experiments. This will establish criteria that can be used to estimate the efficacy of magnetic field flow damping on crystallization of candidate proteins. 3. To demonstrate the understanding of convection damping by high magnetic fields to a class of proteins that is of interest and whose structure is as yet not determined. 4. To compare quantitatively, the quality of the grown crystals with and without a magnetic field. X-ray diffraction techniques will be used for the comparative studies. Prior ambiguities of space-based protein crystallization need clarification. Understanding the fundamentals of a process is critical to major innovations and breakthroughs. Protein crystallization is however a complex process involving the interactions of several physical phenomena such as kinetics, fluid flow, physical chemistry, self-assembly, etc. and is still an empirical science. It holds the promise of the development of next generation designer drugs for and hence it is vigorously pursued by the drug industry. Protein crystallization experiments have flown on more space missions than any other targeted research effort but a comprehensive understanding of the fundamental processes and the beneficial effects of the microgravity environment are not clear. This investigation uses magnetic fields to counter gravitational effects – a terrestrial simulation of microgravity, in order to investigate crystal growth phenomena in reduced gravity. The current working hypothesis is that convection is detrimental to macromolecular crystal growth, although the exact mechanisms are unclear, and yet only 20% of space-grown proteins have shown results better than ground-based studies. This investigation will clarify the impact of fluid flow during the growth process either as a causative factor of some undesirable effects or point to the need to investigate other (non-flow) effects to improve crystal quality. The study adopts a stepping stone approach to the problem by validating theory by simple experiments before moving on to full-fledged protein crystallization. It represents an important contribution in the understanding of the complex crystallization phenomena.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	To date, on this project, a novel, proven technique of obtaining zero-gravity conditions, terrestrially, for protein crystallization conditions has been successfully demonstrated. Protein crystals grown in reduced gravity realize the so-called diffusion limited (no convection) growth conditions. Both solutal buoyancy driven convection (flows due to density differences based on concentration) and sedimentation are reduced significantly, thereby providing a quiescent environment for crystal growth. There is evidence from past missions that some proteins grown in space (~20%) are of better quality than terrestrially grown crystals. Crystallization experiments in space flight are expensive, opportunities are sparse, and are not amenable to a quick-turnaround iterative process that terrestrial experimentation can provide. The new technology developed in this investigation can provide protein crystal growth conditions similar to that realized during space experimentation. The technique is based on using strong magnetic fields to achieve the desired results.
Task Progress:	Several significant accomplishments were recorded in the project in 2005. These are itemized below: 1. A novel technique of magnetic convection control in the above system using a paramagnetic salt (MnCl2.4H2O) was formulated, crystals grown, and flow control theory developed. The importance of this finding is that the crystal growth solution now becomes paramagnetic allowing us to test flow control in protein crystallization using the in-house (MSFC) 5 T superconducting magnet. Experimental verification of the flow control technique and its stability was completed in May 2005. 2. A paper on the above theory and experiments has been submitted to the Journal of Crystal Growth. 3. A new technology disclosure report based on the above findings filed with NASA has been accepted and is undergoing scrutiny and research for a possible Patent application. 4. Second generation flow and diagnostics apparatus to study protein crystal growth convection control in magnetic fields was designed and built with the help of visiting summer faculty to MSFC. It will be used in subsequent testing. 5. Experience from the project was successfully applied to test artificial gravity tissue culture concept for supporting NASA exploration goals. A series of levitation experiments was completed using polymer (PMMA) beads. A review paper on the subject was written and presented at the International Conference on Environment Systems in June 2005. 6. A set of lysozyme crystal growth experiments were completed at Brown University. The primary aim of the experiments was to check the crystal growth cell design, instruments used for diagnostics and the protocol for extended duration crystal growth (several days duration). Power supply problems with the magnet restricted the number tests that could be conducted. A couple of seeded crystals were successfully grown over 2 days of magnet operation. 7. An Advanced X-ray Diffraction Imaging and Scattering Facility for has been

	identified for characterizing protein crystals grown in the MSFC 5 Tesla magnet. The facility ios taergeted to be on line by Feb, 2006. This will allow growth and characterization studies to be completed in-house with little time delay.
Bibliography Type:	Description: (Last Updated: 03/17/2017)
Articles in Peer-reviewed Journals	Ramachandran N, Leslie FW. "Using magnetic fields to control convection during protein crystallization—analysis and validation studies." Journal of Crystal Growth, 2005 Jan 15;274(1-2):297-306. , Jan-2005
Awards	Ramachandran N, Leslie F. "NASA MSFC Director's Commendation for dedicated service and technical contributions and advances in Research – "for the development of a theoretical foundation for the stability criteria of solutal convection of a magnetized fluid and successful verification of the model by experiments," July 2004." Jul-2004
Awards	Ramachandran N, Leslie FW. "NASA Space Act Award and Certificate of Recognition for technical innovation and contribution to NASA, August 2004. " Aug-2004
Papers from Meeting Proceedings	Leslie FW, Ramachandran N. "Terrestrial Microgravity Model and Threshold Gravity Simulation using Magnetic Fields." 35th International Conference on Environment Systems, Rome, Italy, July 11-15, 2005. ICES Paper 2005-01-2765. 2005. , Jul-2005
Presentation	Ramachandran, N., "Using Magnetic Fields for Simulating Reduced Gravity Environments" NASA MSFC Science Directorate Seminar Presentation Sep-2004